SC Algebra 1

Reteach: Intercepts and Intervals

Name: _____

Page 1 — The concepts

1. x- and y-intercepts

The rule

x-intercept: where the graph crosses the _____. There, y = ___.

y-intercept: where the graph crosses the _____. There, x = ___.

x-int → make y = 0. y-int → make x = 0.

Example: 2x + 3y = 12

x-int: 2x + 3(0) = 12 → x = ___ → (___, 0)

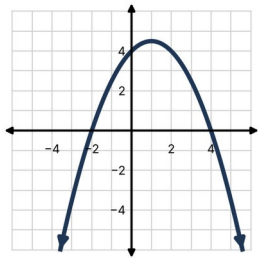
y-int: 2(0) + 3y = 12 → y = ___ → (0, ___)

Example: y = 2x − 6

x-int: 0 = 2x − 6 → x = ___

y-int: y = 2(0) − 6 = ___

From a graph

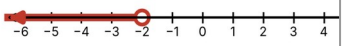
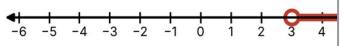


x-int: _____

y-int: _____

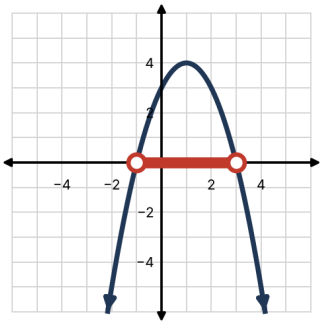
2. What is an interval?

An interval is a _____ of the number line: the x-values between two boundary points, or going on forever in one direction. An open circle means the boundary point is ____ included. **Trick:** the pointy end of < points at the SMALLER number.

LEFT of a number	RIGHT of a number	BETWEEN two numbers	TWO pieces
			
x < −2	**x > 3**	**−1 < x < 4** *smaller number on the left*	**x < −1 and x > 4**

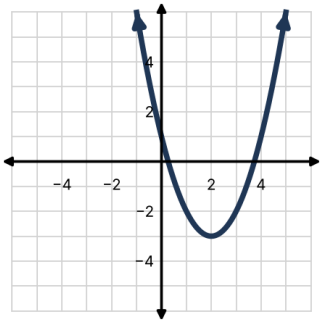
3. Intervals on a graph (always use x-values)

Positive and negative



1. Mark where it crosses the x-axis: x = ___, x = ___
2. ABOVE the x-axis = positive: _____
3. BELOW the x-axis = negative: _____

Increasing and decreasing



1. Find where it turns: x = ___ (use the x-value!)
2. Trace LEFT to RIGHT. Going down? decreasing: _____
3. Going up? increasing: _____

Watch out: × y-values × 5 < x < −2 (smaller number first) × −1 > x > 3 for two pieces (write x < −1 and x > 3)

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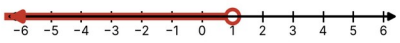
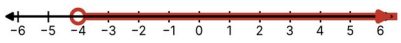
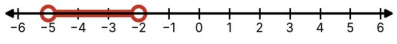
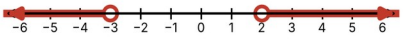
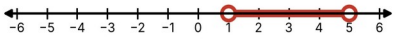
Name: _____

Page 2 — Drill

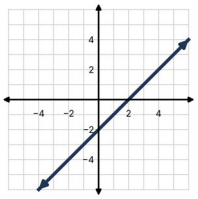
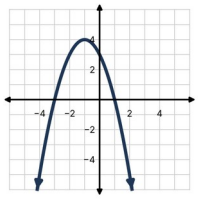
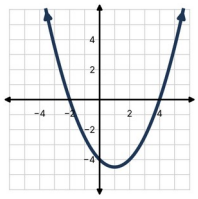
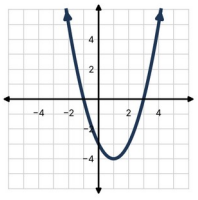
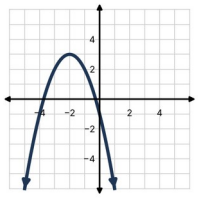
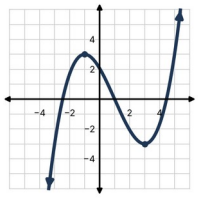
A. Find the x- and y-intercepts.

1. $x + 2y = 8$ x: _____ y: _____	**2.** $3x + y = 9$ x: _____ y: _____	**3.** $4x - 2y = 16$ x: _____ y: _____	**4.** $5x + 2y = -10$ x: _____ y: _____
5. $-3x + 4y = 12$ x: _____ y: _____	**6.** $6x - 5y = 30$ x: _____ y: _____	**7.** $2x - 7y = 14$ x: _____ y: _____	**8.** $-x - 3y = 6$ x: _____ y: _____
9. $y = 3x + 12$ x: _____ y: _____	**10.** $y = -2x + 8$ x: _____ y: _____	**11.** $y = \frac{1}{2}x - 3$ x: _____ y: _____	**12.** $y = -4x - 10$ x: _____ y: _____

B. Write the inequality.

13.  _____	**14.**  _____	**15.**  _____
16.  _____	**17.**  _____	**18.**  _____

C. Positive / negative (top row) and increasing / decreasing (bottom row).

19.  pos: _____ neg: _____	**20.**  pos: _____ neg: _____	**21.**  pos: _____ neg: _____
22.  inc: _____ dec: _____	**23.**  inc: _____ dec: _____	**24.**  inc: _____ dec: _____